Taejun Yoon

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Research Interests

Exploring efficient AI and data systems

- **LLM Inference Optimization:** Improving the throughput and memory efficiency of LLM serving systems (e.g., vLLM) for long-context and agentic workloads, with a focus on KV-cache management
- **Sparse Attention:** Designing retrieval-based sparse attention with KV-cache offloading to host memory, enabling larger batch sizes for long-context decoding
- **Cloud Data Systems:** Designing and evaluating caching for direct-access cloud OLAP over object storage

Education

POSTECH, Dept. of Computer Science and Engineering — Mar. 2024 – Present
M.S./Ph.D. Integrated Program
Data Systems Lab
Advisor: Prof. Wook-Shin Han

POSTECH, Dept. of Computer Science and Engineering — Feb. 2020 – Feb. 2024
B.S. in Computer Science and Engineering
GPA: 4.07 / 4.3

Professional Experiences

Data Systems Lab, POSTECH, Korea — Jun 2022 – Feb 2024
Research Intern

- Conducted research on database recovery
- Implemented efficient data loading for a distributed database system

Publications

1. **Taejun Yoon**[*], Taesung Lee[*], Joo Yeon Lee, Ki Hong Kim, Byungchul Tak, Wook-Shin Han. ([*]equal contribution) *Cache Replacement in Direct-Access Cloud OLAP—An Experimental Comparison and Practical Guidance [Experiments & Analysis].* SIGMOD 2027.

Projects

Cache Replacement for Direct-Access Cloud OLAP — Mar 2024 – Jun 2026
Industry Collaboration with SAP Labs Korea

- **Problem:** Lakehouse query engines that read data directly from cloud object storage rely on local caching, but the object and byte hit ratios used to evaluate replacement policies poorly reflect actual query latency
- **Contribution:** Proposed UHR (Unified Hit Ratio), a latency-aware cache metric, and experimentally compared 22+ replacement policies on SAP HANA Cloud with analytical workloads, deriving practical guidance for policy selection
- **Outcome:** Accepted at SIGMOD 2027 (Experiments & Analysis)

Technical Skills

Programming Languages	C/C++ (Advanced), Python (Advanced), SQL (Advanced)
Databases	PostgreSQL, DuckDB
Systems & Build Tools	Linux, Git, Docker, CMake
ML/LLM	PyTorch, vLLM, OpenAI API

Languages

Korean	Native
English	Professional

Teaching Experiences

- **Teaching Assistant**, CSED321: Programming Languages, POSTECH — Spring 2026
- **Teaching Assistant**, CSED421: Database Systems, POSTECH — 2024

References

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